

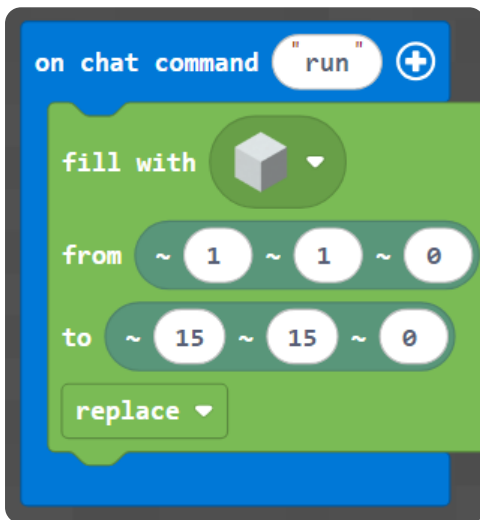
M003 Week 4 — English Worksheet (Beginner)

Topic: Pixel Art 2 — Cat & Tree (x, y) · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 30 minutes

This week you copy **two more pictures** — a **cat** and a **tree**. Every block sits at a spot named by **two numbers** — (x, y). x is how far **across**, y is how far **up**. You read the picture and place each block at its (x, y).

First, build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Stand on the red block every time you run.** If you move, the picture moves too and the blocks land in the wrong place. Same spot every time = a clean picture.

1 · Predict 🧠

Read the steps. Before you imagine it happening, circle or write your answer.

```
place brown block at (8, 1)
place brown block at (8, 2)
place brown block at (8, 3)
```

Do these blocks make a line going up? Circle one: across · up

Which number stays the same — x or y? (This is the tree trunk.)


```
place black block at (3, 15)
place black block at (4, 15)
```

These share $y = 15$ and x grows. Do they go across or up? Circle one: across · up

2 · Find the Difference 🐛

Each pair shows the **correct** line first, then a **broken** one. Circle what's different and write one short sentence about the bug.

Pair A — The tree trunk should grow **up** at $x = 8$: (8, 1), (8, 2), (8, 3).

```
# correct
place brown block at (8, 1)
place brown block at (8, 2)
place brown block at (8, 3)
```

```
# broken
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
```

What is wrong? Where do all three blocks land?

Pair B — The trunk should grow **up** at $x = 8$.

```
# correct
place brown block at (8, 2)
place brown block at (8, 3)
```

```
# broken
place brown block at (2, 8)
place brown block at (3, 8)
```

What is wrong? Which two numbers got swapped?

3 · Fill the Gap

To grow the trunk **up**, the **second** number (y) has to grow. Fill in the missing numbers so the trunk goes up.

```
place brown block at (8, 1)
place brown block at (8, __)
place brown block at (8, __)
```

Word bank for the blanks: 2 · 3

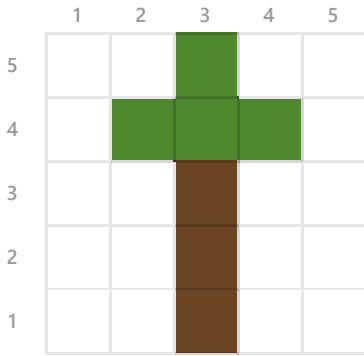
Write the two missing numbers:

4 · Tell Me What You Built 📷

Stand on your **home spot** (the red block).

Warm up first. Copy this small tree onto your wall. Read each square's (x, y) and place that color there.

 Small tree



Now pick one big picture — the 🐱 cat or the 🌳 tree from the main worksheet grid — and copy it the same way. Read each colored square and place that color at its (x, y).

When you finish, send a photo of your picture, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

I built the ...

The lowest block in my picture is at $(x = \dots, y = \dots)$.

One tricky moment was when ...

5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while you show your picture on the wall. Talk like you are teaching a friend. Try to use these words: **x, y, across, up, home spot**.

1. Show your home spot (the red block) and say why you stand on it.
2. Show your finished picture.
3. Point at one block and read its coordinate: "this one is at x ..., y ...".

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.